Limiti di funzioni

Siano $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, $\ell \in \mathbb{R}^*$ (→ $\mathbb{R} \cup \{-\infty, +\infty\}$) si dice

limite di $f(x)$ per $x \to x_0$

$$\lim_{x \to x_0} f(x) = \ell \qquad \left(f(x) \to \ell \text{ per } x \to x_0 \right)$$

se

$\forall$ intorno V di ℓ, $f(x) \in V$ definitivamente per $x \to x_0$

N.B. $$\lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x) = \ell$$

Def.

Una funzione $f: X \to \mathbb{R}$ si dice <u>infinitesima</u> per $x \to x_0$ se

$$\lim_{x \to x_0} f(x) = 0$$

$$\left(f(x) \to 0 \quad \text{per } x \to x_0 \right)$$

$$\left(f(x) = o(1) \quad \text{per } x \to x_0 \right)$$

Una funzione $f: X \to \mathbb{R}$ si dice <u>infinita</u> per $x \to x_0$

$$\lim_{x \to x_0} f(x) = \pm \infty$$

$$\left(f(x) \to \pm \infty \quad \text{per } x \to x_0 \right)$$

Esempio

$f(x) = \sqrt{x}$ per $x \to 0$ $\Rightarrow$ f è infinitesima

$f(x) = \frac{1}{x}$ per $x \to 0$ $\Rightarrow$ f non è infinita né infinitesima

$f(x) = \frac{1}{x}$ per $x \to 0^+$ (per $x \to 0^-$) $\Rightarrow$ f è infinita

Simboli di Landau

Date due funzioni f e g, definite in un intorno U di $x_0 \in \mathbb{R}^*$ si dice f è un "o piccolo" di $g(x)$ se

↳ notazione di Landau

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0$$

$\left(\frac{f(x)}{g(x)} \to 0 \text{ per } x \to x_0 \right)$

se $g(x) \neq 0$ definitivamente per $x \to x_0$.

N.B.
$f(x)$ è infinitamente + piccola di $g(x)$

Esempi

(i) $x^2 = o(kx^3)$ per $x \to \infty$

$f(x) = x^2$

$g(x) = x^3$

$$\lim_{x\to\infty} \frac{x^2}{x^3} = \lim_{x\to\infty} \frac{1}{x} = 0$$



(ii) $x^3 = o(x^2)$ per $x \to 0$

$$\lim_{x\to 0} \frac{x^3}{x^2} = \lim_{x\to 0} x = 0$$

(iii) $x^2 = o(2^x)$ per $x \to \infty$

(iv) $e^x = 1 + x + o(x)$ per $x \to 0$

$f(x) = e^x - 1 - x$

$g(x) = x$

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x} = \lim_{x\to 0} \frac{e^x - 1}{x} - \frac{x}{x} =$$

$$\lim_{x\to 0} \frac{e^x - 1}{x} - 1 = 1 - 1 = 0$$

$\Rightarrow$

$f(x) = o(g(x))$

Operazioni con simboli Landau

per $x \to 0^{\pm}$ e per $x \to \pm\infty$, $\alpha, \beta \in \mathbb{R}$

i) $o(Kx^{\alpha}) = K\,o(x^{\alpha}) = o(x^{\alpha}) \qquad \forall K \neq 0$

ii) $o(x^{\alpha}) + o(x^{\alpha}) = -o(x^{\alpha})$

$o(x^{\alpha}) - o(x^{\alpha}) = o(x^{\alpha})$

iii) $o(x^{\alpha}) \cdot x^{\beta} = o(x^{\alpha+\beta})$

iv) $o(x^\alpha) \cdot o(x^\beta) = o(x^{\alpha+\beta})$

$f(x) = o(x^\alpha)$

$g(x) = o(x^\beta)$

$$\lim_{\substack{x\to 0^\pm \\ (x\to\pm\infty)}} \frac{f(x)\cdot g(x)}{x^{\alpha+\beta}} = \lim_{\substack{x\to 0^\pm \\ (x\to\pm\infty)}} \frac{f(x)}{x^\alpha} \cdot \frac{g(x)}{x^\beta} = 0$$

$\frac{f(x)}{x^\alpha} \to o(x^\alpha)$

$\frac{g(x)}{x^\beta} \to o(x^\beta)$

v) $o(o(x^\alpha)) = o(x^\alpha)$

vi) per $x \to 0^\pm$

① $x^\alpha = o(x^\beta)$ se $\alpha > \beta$

② $x^\alpha = o(x^\beta)$ se $\alpha < \beta$

per $x \to \pm\infty$

① $\lim_{x\to 0} \frac{x^\alpha}{x^\beta} = 0$

$\lim_{x\to\pm\infty} \frac{x^\alpha}{x^\beta} = 0$

vii) $o(x^\alpha) \pm o(x^\beta) = o(x^\alpha)$ per $x \to 0$ se $\alpha < \beta$

$o(x^\alpha) \pm o(x^\beta) = o(x^\beta)$ per $x \to \infty$

Limiti notevoli (riformulazione con Landau)

⚠ per $x \to 0$

$$e^x = 1 + x + o(x)$$

$$a^x = 1 + x \log a + o(x) \qquad \forall a > 0$$

$$\log(1+x) = x + o(x) \qquad \lim_{x\to 0} \frac{\log(1+x)}{x} - 1 = 1 - 1 = 0$$

$$(1+x)^\alpha = 1 + \alpha x + o(x) \qquad \forall \alpha \in \mathbb{R}$$

$$\sin x = x + o(x)$$
$$\tan x = x + o(x)$$
$$\cos x = 1 - \frac{1}{2}x^2 + o(x^2)$$

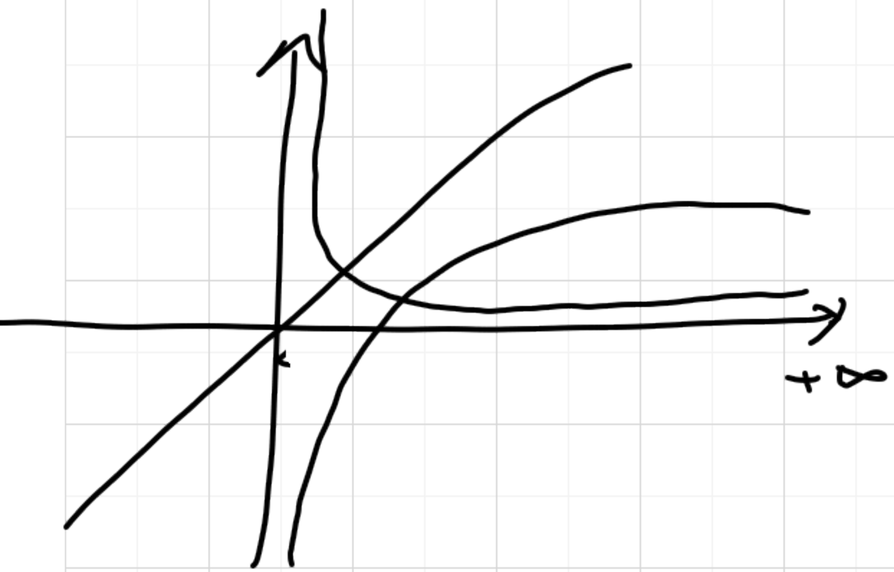
Esse

$$\lim_{x\to\infty} \frac{\log(3x^6-5)}{\sqrt[8]{x+1}} =$$

$$= \lim_{x\to\infty} \frac{\log\left(x^6\left(3-\frac{5}{x^6}\right)\right)}{\sqrt[8]{x\left(1+\frac{1}{x}\right)}} =$$

$$= \lim_{x\to\infty} \frac{\log x^6 + \log\left(3-\frac{5}{x^6}\right)}{\sqrt[8]{x\left(1+\frac{1}{x}\right)}} \sim \frac{6\log x + \log(3+o(1))}{\sqrt[8]{x(1+o(1))}}$$

$$o(-5) = -5\,o(1)$$



$= o(1)$

$$\frac{)}{} = \frac{6 \log x}{x^{\frac{1}{8}}} \to 0$$

Ese

$$\underbrace{\cos(x^2) - \sqrt{1-x^4} + x^5}_{f(x)} = o\underbrace{(x^4)}_{g(x)=x^4} \quad \text{per } x \to 0$$

È vera?

$$\lim_{x\to 0} \frac{\cos(x^2) - \sqrt{1-x^4} + x^5}{x^4} = \lim_{x\to 0} \frac{x^5 + \cos(x^2) - 1 + 1 - \sqrt{1-x^4}}{x^4}$$

$$\sim \frac{-\frac{1}{2}(x^2)^2 - \frac{1}{2}(-x^4) + x^5}{x^4} = \frac{x^5}{x^4} = 0$$

$\cos x - 1 = -\frac{1}{2}x^2 + o(x^2)$

$(1+x)^\alpha - 1 = \alpha x + o(x)$

TRACCIA SENZA x^5

$$\sim \frac{o((x^2)^2) - o(x^4)}{x^4} = \frac{o(x^4)}{x^4} = 0$$

$)^{\frac{1}{2}}$

Ese

$$\lim_{x\to+\infty} \sqrt{x^2+1} - \sqrt{x} =$$

$$= \lim_{x\to+\infty} \sqrt{x^2\left(1+\frac{1}{x^2}\right)} - \sqrt{\frac{x^2}{x}} =$$

$$= \lim_{x\to+\infty} x\sqrt{1+\frac{1}{x^2}} - x\sqrt{\frac{1}{x}} = \lim_{x\to+\infty} x\left(\sqrt{1+\frac{1}{x^2}} - \sqrt{\frac{1}{x}}\right) =$$

$\frac{1}{x^2} \to 0$, $\frac{1}{x} \to 0$, $\to 1$

Esempio

$$\lim_{x\to+\infty} \frac{3x^3 + 3^x + \log x}{\log x + 2^{2x} + x^5} =$$

$$= \lim_{x\to+\infty} \frac{3^x}{2^{2x}} = \lim_{x\to+\infty} \left(\frac{3}{4}\right)^x = 0$$

$\left(\frac{3}{4}\right) < 1$

$: +\infty$

$$\lim_{x\to+\infty} b^x = \begin{cases} +\infty & b > 1 \\ 1 & b = 1 \\ 0^+ & \text{se } b < 1 \end{cases}$$

Esempio

$$\lim_{x\to+\infty} \sqrt[x]{x^2} = \lim_{x\to+\infty} x^{\left(\frac{2}{x}\right)} \to 0$$

∞^0

$= 1$

SBAGLIATO

$$\log\left(x^{\frac{2}{x}}\right) \quad \frac{2}{x}$$

$$\frac{2}{x}\log x \qquad \frac{2\log x}{x}$$

per $x\to\infty$ $\frac{\infty}{\infty}$

$$= e^{\log\left(x^{\frac{2}{x}}\right)} = e^{\frac{2}{x}\log x} = e^{\frac{2\log x}{x}} = e^0 = 1$$

Ese 4.7.1 a

$$\lim_{x\to\infty} \frac{x+1}{\sqrt{x}} - \frac{x}{\sqrt{x+1}} =$$

$$= \lim_{x\to\infty} \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} - \frac{x}{\sqrt{x\left(1+\frac{1}{x}\right)}} =$$

$$= \lim_{x\to\infty} \sqrt{x} + \frac{1}{\sqrt{x}} - \frac{x}{\sqrt{x}\sqrt{1+\frac{1}{x}}} =$$

$$= \lim_{x\to\infty} \sqrt{x} + \frac{1}{\sqrt{x}} - \sqrt{x}\cdot\frac{1}{\left(1+\frac{1}{x}\right)^{\frac{1}{2}}} =$$

$$= \lim_{x\to\infty} \sqrt{x} + \frac{1}{\sqrt{x}} - \sqrt{x}\cdot\left(1+\frac{1}{x}\right)^{-\frac{1}{2}} =$$

$$= \lim_{x\to\infty} \sqrt{x} + \frac{1}{\sqrt{x}} - \sqrt{x}\left(1+\frac{1}{2}\cdot\frac{1}{x} + o\left(\frac{1}{x}\right)\right) =$$

$$= \lim_{x\to\infty} \sqrt{x} + \frac{1}{\sqrt{x}} - \sqrt{x} - \frac{\sqrt{x}}{2\sqrt{x}\cdot\sqrt{x}} - x^{\frac{1}{2}}\cdot o\left(x^{-1}\right) =$$

$$= \lim_{x\to\infty} \frac{1}{\sqrt{x}} - \frac{1}{2\sqrt{x}} + o\left(x^{-1+\frac{1}{2}}\right) = \lim_{x\to\infty} \frac{1}{2\sqrt{x}} + o\left(x^{-\frac{1}{2}}\right)$$



$$\lim_{x\to\infty} \frac{x^2}{\sqrt{x^2+3x+1}} - x$$

$$\lim_{x\to 0} \log 3^{-\frac{1}{x}} - \frac{1}{x}\log\left(\left(x+\frac{1}{3}\right)(1-2x)\right)$$

$$\lim_{x\to 0^+} \frac{\sqrt[3]{x\sin^2(2x)}\cdot\left(\sqrt{x}-1\right)}{e^x-\cos(3x)}$$

$=0$

Esse $\lim_{x \to 0^+} \dfrac{\sin x}{\cos(\sqrt{x}) - 1} \sim \dfrac{x + o(x)}{-\frac{1}{2}(\sqrt{x})^2 + o(x)} = -2$

Esse $(-x^{\alpha}) \left(\dfrac{x}{\sqrt{1-x}} - \dfrac{(1-x)^{\frac{3}{2}}}{x} \right)$ per $x \to -\infty$

$(-x^{\alpha}) \left(\dfrac{x^2 - (1-x)^{\frac{3}{2}} \cdot (1-x)^{\frac{1}{2}}}{x(\sqrt{1-x})} \right) =$

$$= (-x)^{\alpha} \cdot \frac{x^2 - (1-x)^2}{x\left(\sqrt{1-x}\right)} =$$

$$= \left(-x^{\alpha}\right) \cdot \frac{\cancel{x^2} - \cancel{x^2} - 1 + 2x}{x\left(\sqrt{1-x}\right)} =$$

$$= \left(-x^{\alpha}\right) \cdot \frac{\cancel{x}\left(2 - \frac{1}{x}\right)}{\cancel{x}\left(\sqrt{1-x}\right)} = \left(-x^{\alpha}\right) \cdot \frac{2 - \frac{1}{x}}{\sqrt{x}\left(\sqrt{1-\frac{1}{x}}\right)} =$$

$$\sim -2x^{\alpha} \cdot x^{-\frac{1}{2}}$$

$\frac{1}{x} \to 0$, $\sqrt{1-\frac{1}{x}} \to 1$

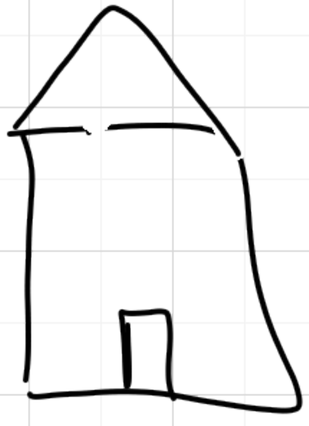
$$= -2 x^{\alpha - \frac{1}{2}}$$

$$\Rightarrow \begin{cases} -\infty \\ -2 \\ 0 \end{cases}$$

$\alpha > \frac{1}{2}$ $\boxed{\alpha - \frac{1}{2} \geq 0}$

$\alpha = \frac{1}{2}$

$\alpha < \frac{1}{2}$



$$\lim_{x \to \infty} \left(\frac{1}{2} + e^{\alpha x} \right)^{\log\left(-\frac{1}{x}\right)}$$

per $x \to -\infty$

Se $\alpha \geq 0$